

Value Chain Analysis and AI Integration Strategy for Abbott Diabetes Care

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Abbott Diabetes Care (ADC), a division of Abbott Laboratories, is a global leader in continuous glucose monitoring (CGM) technology. This report analyzes ADC's value chain to identify the most critical activities that create value and explores how artificial intelligence (AI) can enhance these activities. A comparative analysis with a primary competitor is included to assess relative AI integration potential, followed by strategic recommendations.

Porter's value chain framework divides firm activities into primary and support categories (Kennedy, 2010). ADC's operations and technology development are particularly critical to its CGM product line.

Primary Activities

Product Development and Operations: ADC's scientific development of CGM systems—including sensor engineering, algorithm calibration, and clinical validation—is central to its value creation. These activities directly impact product quality, regulatory compliance, and differentiation (Magretta, 2011).

Manufacturing: ADC manufactures many CGM components in-house, including sensors and readers. This allows for tight quality control, cost efficiency, and scalability (Abbott, 2022).

Outbound Logistics: Distribution of FreeStyle Libre products through pharmacies, hospitals, and direct-to-consumer channels.

Marketing and Sales: Includes digital and print advertising, branded websites (e.g., freestylelibre.abbott), educational materials for healthcare providers (HCPs), and promotional tools such as the "Add to Wallet" feature for free trial cards.

Service: Customer support via phone and chat, warranty handling, and technical assistance for device setup and app integration.

Support Activities

Technology Development: Scientific research and clinical testing for CGM accuracy, sensor innovation, and algorithm development. Also includes software development for patient-facing tools and mobile apps.

Information Technology (IT): Split into three functional areas:

- *Enterprise IT:* Supports internal systems and infrastructure utilized by Abbott Employees.

- *Product IT*: Develops consumer-facing digital tools and mobile applications utilized by CGM customers.

- *Digital Delivery (Marketing IT)*: Develops consumer-facing websites and e-commerce platforms.

Human Resource Management: Recruitment and retention of talent in biotech, software engineering, regulatory affairs, and customer support.

Procurement: Sourcing of raw materials, components, and cloud infrastructure; negotiation of vendor contracts for data hosting and analytics platforms.

Firm Infrastructure: Legal, finance, compliance, and global regulatory coordination.

The linkages between product development, manufacturing, and service are particularly strong. Scientific innovation drives product quality, which in turn affects customer satisfaction and brand trust—key elements of ADC’s differentiation strategy (Magretta, 2011).

Potential for AI to Enhance Activities

AI can enhance both primary and support activities when aligned with strategic goals (Kwong, 2025). For Abbott Diabetes Care (ADC), four activities stand out as high-impact opportunities for AI integration:

Product Development and Operations: AI can accelerate sensor calibration, improve glucose trend prediction, and personalize health insights. These capabilities support ADC’s differentiation strategy by enhancing product performance and user experience (Magretta, 2011).

Manufacturing: AI-driven predictive maintenance and quality control can reduce downtime and defects, improving cost efficiency and throughput (Edwards, 2014).

Marketing and Sales: AI can optimize campaign targeting and customer segmentation, increasing engagement and conversion rates (Harvard Business School Press, 2006).

Customer Support (Service): AI-powered tools such as small language models (SLMs) can assist customer service agents in real time by retrieving accurate responses from a vast knowledge base, including over 700 FAQs on the FreeStyle Libre U.S. website. This can reduce average handling time, improve first-call resolution, and enhance customer satisfaction. As Kwong (2025) notes, AI can streamline service delivery by

aligning support activities with customer needs, especially when integrated with internal systems and reviewed by human agents.

Competitive Analysis:

Dexcom, a leading CGM competitor, also invests in R&D and digital integration. In December 2024, Dexcom announced the launch of a proprietary generative AI platform built on Google Cloud's Vertex AI and Gemini models. This platform is designed to deliver personalized insights to users and support future product innovation (DexCom, Inc., 2024).

Dexcom's strategic direction includes expanding into Type 2 and wellness markets, supported by partnerships with ŌURA Health and CMS. Its CEO has emphasized the role of AI in food logging and consumer engagement (AInvest, 2025). However, Dexcom relies more heavily on external partnerships for manufacturing and cloud infrastructure, which may limit its control over quality and compliance compared to Abbott's vertically integrated model (SWOTAnalysis.com, 2025).

Abbott's in-house manufacturing and proprietary data access give it an edge in deploying AI internally, especially in operations, product development, and customer support (Kennedy, 2010).

Recommendations

Abbott's *Code of Business Conduct* emphasizes the company's commitment to "Caring," defined as prioritizing the health and well-being of patients and consumers through ethical, responsive, and innovative practices (Abbott Laboratories, 2023). In keeping with this value—and with Abbott's cautious approach to emerging technologies—these recommendations focus on conservative, low-risk areas of AI integration that enhance operational efficiency and customer experience without compromising regulatory compliance, data privacy, or human oversight.

- **Pilot AI-powered agent assist tools** in customer support to improve service consistency, reduce response times, and increase customer satisfaction.
- **Implement AI-driven feedback analysis** to identify patterns in customer complaints, usability issues, and support gaps. This can inform updates to FAQs, training protocols, and even product design. By

linking customer sentiment with operational data, ADC can proactively improve the user experience and reduce support burden.

I am certain there are more opportunities for implementing AI, but want to focus on initial pilots that support scalable, measurable improvements across the value chain. These recommendations reflect a values-driven approach to AI adoption—one that supports Abbott's strategic goals while honoring its commitment to patient safety, regulatory integrity, and responsible innovation.

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